

A Certified Framework for Deterministic Navigation in Higher-Genus p -Isogeny Graphs

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Abstract

We present a deterministic framework for navigating p -isogeny graphs of genus $g \geq 2$, addressing the lack of canonical and auditable primitives in higher dimensions. The framework integrates two components: the Certified p -Isogeny Step (PICS) and a Non-Decomposition Certificate (ND). PICS constructs the unique Frobenius-compatible inseparable isogeny by extracting kernel directions from Hasse–Witt invariants and differential subresultant profiles, thereby eliminating randomized kernel selection. Complementarily, ND serves as an algebraic filter that rejects Jacobians compatible with product decompositions by enforcing cyclicity in the associated differential operator module. We prove that the rejection density scales asymptotically as $O(p^{-1})$. Experimental validation using a C-based backend over 256-bit prime fields demonstrates that the certification logic incurs a relative overhead of less than 0.2% compared to the mandatory Hasse–Witt computation. By enforcing strict determinism and structural safety, the resulting transition unit provides a verifiable primitive for auditable parameter generation and isogeny-based time-lock puzzles.

Keywords: certified isogenies ; p -isogenies ; Frobenius kernels ; Hasse–Witt matrices ; differential subresultants ; verifiable delay functions

1 Introduction

Isogenies between abelian varieties have emerged as a central substrate for post-quantum cryptography. While the elliptic curve case ($g = 1$) has reached a level of algorithmic maturity, the landscape of higher-dimensional abelian varieties ($g \geq 2$) remains a frontier characterized by both high potential and significant structural risk. Jacobians of higher-genus curves offer exponentially denser isogeny graphs [15, 11] and richer endomorphism rings,

but they simultaneously introduce algebraic complexities—and potential vulnerabilities—that are absent in dimension one.

1.1 The Post-SIDH Landscape and Structural Hygiene

The cryptographic paradigm for isogenies shifted fundamentally following the cryptanalysis of SIDH by Castryck and Decru [3], with subsequent refinements [16, 12]. These attacks exploited the presence of auxiliary torsion information combined with the ability to map the target abelian variety into a product of lower-dimensional varieties (specifically, a decomposable Jacobian $J \cong E_1 \times E_2$).

This episode highlighted a critical lesson: *structural safety is a prerequisite for security*. In higher dimensions, the moduli space of curves $\mathcal{M}_g$ maps into the moduli space of abelian varieties $\mathcal{A}_g$ via the Torelli morphism. The image contains a dense union of “special” loci where Jacobians decompose or possess exceptional endomorphisms. Navigating these graphs without explicit algebraic guards poses a non-trivial risk: a random walk or a parameter generation procedure might inadvertently drift into a decomposable locus, potentially reducing the effective security level or facilitating projection-based attacks. Thus, the challenge in the post-SIDH era is not merely to compute isogenies, but to navigate the graph with *certified structural hygiene*, ensuring that the protocol remains strictly within the absolutely simple locus.

1.2 The Deficit of Canonical Primitives

Despite these risks, the algorithmic toolkit for $g \geq 2$ lacks the canonical primitives available for elliptic curves. While recent work by Dang [5] successfully reformulated the Richelot $(2, 2)$ -isogeny using differential subresultants to achieve a derivative-free and guarded evaluation, that construction remains tied to the specific modular geometry of degree-2 isogenies. Extending such formula-based approaches to arbitrary degrees and general navigation encounters two fundamental obstacles:

1. **Absence of Canonical Navigation for Large p :** For elliptic curves, the p -power Frobenius provides a unique, inseparable direction. In higher genus, extracting a deterministic kernel for a $(p, \dots, p)$ -isogeny—one that is uniquely dictated by the curve’s moduli point rather than by external choices—is algebraically subtle. Existing constructions relying on factorization patterns (like Richelot) [4] face a combinatorial explosion when generalized to arbitrary primes p .
2. **Lack of Auditability:** Current methods for walking isogeny graphs often depend on randomized kernel selection. This introduces “surveyor discretion”: an external auditor cannot verify whether a public parameter set was generated honestly or selected via “grinding” to

embed a hidden weakness (e.g., a secret decomposition). In trusted setup scenarios, this opacity is unacceptable.

1.3 Our Approach: A Certified, Deterministic Framework

In this work, we elevate the insight of [5] from a specific algorithmic formulation to a general invariant-theoretic principle. We introduce a certification-oriented framework for deterministic navigation in higher-genus p -isogeny graphs. We construct a **Certified Transition Unit** that acts as a verifiable atomic step, ensuring both canonical evolution and structural safety.

Our approach moves beyond the formula-based paradigms of genus-2 literature. Instead, we employ invariant theory—specifically **Frobenius–Cartier constraints** and **Differential Subresultant (PRS) profiles**—to achieve two guarantees:

- (i) **Deterministic Kernel Extraction (PICS):** We define a primitive that recovers the unique Frobenius-compatible kernel direction directly from the Hasse–Witt and differential data of the curve. This eliminates all auxiliary choices, making the transition mathematically rigid and fully reproducible.
- (ii) **Algebraic Safety Certification (ND):** We introduce a *Non-Decomposition Certificate* (ND) that acts as a conservative filter. By enforcing cyclicity in the operator module associated with the curve’s differential invariants, ND rejects instances compatible with non-trivial product decompositions.

Scope of Application. This framework avoids the optimization-at-all-costs paradigm of key exchange primitives like CSIDH. The inherent cost of determining the canonical Frobenius direction makes individual steps computationally intensive ($10^1 \sim 10^2$ ms). Consequently, our framework is not designed for high-throughput encryption but provides the foundational “safe corridor” for protocols that **prioritize auditability and strict determinism** over raw execution speed. Primary applications include *deterministic parameter generation*, *transparent trusted setups*, and *isogeny-based time-lock puzzles* (VDFs), where the computational cost acts as a feature ensuring sequential security.

1.4 Our Contributions

We formalize the problem of structural navigation in higher-genus isogeny graphs and provide a complete algebraic solution. Our main contributions are:

1. **A Deterministic, Frobenius-Compatible Primitive (PICS).**
We construct the *Certified p -Isogeny Step* (PICS), an algorithm that

extracts the unique inseparable isogeny direction mandated by the Dieudonné module structure [13, 14] of the Jacobian. Unlike genus-specific formulas that rely on randomized kernel selection or factorization patterns, PICS is driven by the Hasse–Witt defect and differential subresultant profiles. We prove that this procedure is canonical and succeeds generically (Theorem 3).

2. **A Sound Non-Decomposition Certificate (ND).** To enforce structural safety, we introduce the *Non-Decomposition Certificate* (ND). This is a computable algebraic filter that rejects Jacobians compatible with a non-trivial product decomposition. By linking the single-drop behavior of differential PRS profiles to the cyclicity of the underlying operator module, we prove that ND acts as a sound one-sided guard against decomposable “weak keys” (Theorem 7). We further show that the rejection density is asymptotically negligible, $O(1/p)$ (Proposition 9).
3. **Auditable Navigation Framework.** We combine PICS and ND into a unified transition unit $\mathcal{U}$. We prove that iterated application of $\mathcal{U}$ yields a deterministic trajectory that is fully determined by the starting seed and certifiably remains within the absolutely simple locus (Theorem 11). This provides the first genus-independent mechanism for *auditable parameter generation*, eliminating the “surveyor discretion” problem inherent in random-walk approaches.
4. **Verification-Centric Validation.** We provide a reference implementation and rigorous benchmarking. Using a C-based backend (FLINT) on cryptographic parameters ($p \approx 2^{256}$), we demonstrate that the structural safety checks incur a latency of only $\approx 19.67\mu s$ per step (relative overhead $< 0.2\%$). Furthermore, extensive statistical sampling confirms the asymptotic sparsity of the rejection locus, which scales as $O(p^{-1})$.

1.5 Related Work

Our work occupies the intersection of algorithmic number theory, arithmetic geometry, and post-quantum cryptography.

Explicit Isogeny Formulas. In genus $g = 2$, the computation of $(2, 2)$ -isogenies is classical, dating back to Richelot, and has been refined for cryptographic use by Smith, Costello, and others [8, 4]. Very recently, theta-model approaches have also been revisited [6]. Recent work by Dang [5] successfully reformulated the Richelot step using differential subresultants to achieve a derivative-free, guarded evaluation. This demonstrated the power of subresultants to detect geometric degeneracy without explicit factorization. However, such methods are intrinsically tied to specific modular geometric

configurations (e.g., intersections of quadrics) and do not generalize to the arbitrary-degree, arbitrary-genus setting required for navigating large p -isogeny graphs. Our framework replaces these ad-hoc geometric formulas with a uniform Dieudonné-theoretic approach [13].

Frobenius and Cartier–Manin Theory. The theoretical interaction between the Hasse–Witt matrix and the p -rank of Jacobians is foundational, established by Hasse, Witt, Manin, and Yui [13]. Algorithmic aspects have been treated by Harvey and Sutherland [10] (for efficient point counting). We repurpose these tools: rather than using them for counting points or determining zeta functions, we use the underlying differential operator module to drive *constructive* isogeny steps.

Structural Attacks and Decomposability. The vulnerability of isogeny protocols to structural auxiliary information was demonstrated by the attacks on SIDH [3, 16]. These attacks leverage the ability to project computations onto lower-dimensional factors of a split Jacobian. While our setting (pure p -isogenies) differs from the torsion-point setting of SIDH, the “risk of decomposability” remains a critical concern for parameter generation. Our ND certificate provides an explicit, computable countermeasure to ensure that generated instances do not fall into these structurally weak loci.

2 Background

This section establishes the requisite algebraic foundations. We recall the Dieudonné-theoretic interplay between the Hasse–Witt matrix and the structure of p -torsion group schemes, and we define the differential subresultant profiles that will serve as our primary certification instrument. Throughout, let $\mathbb{F}_q$ be a finite field of characteristic $p > 2$.

2.1 Hyperelliptic Curves and Jacobians

We work with the smooth projective hyperelliptic curve C associated to the affine model given by

$$C : y^2 = f(x), \quad f \in \mathbb{F}_q[x], \quad \deg f \in \{2g + 1, 2g + 2\},$$

where f is squarefree. Let $J(C)$ denote the Jacobian variety. The space of holomorphic differentials $H^0(C, \Omega^1)$ admits a standard basis $\omega_i = x^{i-1}dx/y$ for $1 \leq i \leq g$.

2.2 Frobenius, Cartier, and Hasse–Witt Invariants

Let Φ_p be the absolute p -power Frobenius endomorphism on C and $J(C)$. The action of Frobenius on the cohomology group $H^1(C, \mathcal{O}_C)$ is induced by the p -linear Cartier operator $\mathcal{C}$ acting on $H^0(C, \Omega^1)$. With respect to the

basis (ω_i) , the *Hasse–Witt matrix* $H \in \text{Mat}_{g \times g}(k)$ is the coefficient matrix of $\mathcal{C}$:

$$\mathcal{C}(\omega_j) = \sum_{i=1}^g H_{ij} \omega_i.$$

The p -rank of $J(C)$, denoted by a , is given by $a = \text{rank}(H)$. The integer $d = g - a$ is the *Hasse–Witt defect*. The structure of the p -torsion subgroup scheme $J(C)[p]$ is governed by its associated Dieudonné module [13, 14]. Crucially for our framework, the Hasse–Witt matrix (and its higher iterates) encodes the constraints on the existence of Frobenius-stable subgroups within $J(C)[p]$.

2.3 Differential Subresultants and Operator Modules

To obtain algebraically verifiable certificates without full point counting, we utilize Hasse derivatives $D^{(e)}$ defined by the Taylor expansion $f(x + T) = \sum D^{(e)} f(x) T^e$ [17, 19]. We focus on two differential pairs:

$$(f, D^{(1)} f) \quad \text{and} \quad (f, D^{(p)} f).$$

These pairs encode the linearization of the curve’s geometry with respect to the tangent bundle and the Frobenius action, respectively.

Subresultant PRS. Let $\text{Sres}_k(P, Q)$ denote the k -th subresultant polynomial of $P, Q \in \mathbb{F}_q[x]$. The sequence of degrees of non-vanishing subresultants (the *Degree Profile*) reflects the structure of the greatest common divisor. More subtly, over the ring of differential operators, this profile encodes the invariant factors of the module associated to the pair [2].

The Determinantal Profile of Cyclicity. A central invariant in our certification logic is the *single-drop profile*. We say that a differential pair exhibits a *single-drop profile* at index j^* if the subresultant degree profile contains exactly one non-generic degree drop (or jump) occurring at j^* . Algebraically, this profile corresponds to the associated operator module being *cyclic* (possessing a single non-trivial invariant factor in its Smith Normal Form).

Linking Algebra to Safety. In the context of Jacobians:

- A generic, absolutely simple Jacobian typically induces a cyclic operator module (single-drop).
- A decomposable Jacobian ($J \sim A_1 \times A_2$) generically induces a split module structure, which manifests as a non-cyclic profile (multiple drops or rank deficiencies).

This connection allows us to use the subresultant profile as a computable proxy for structural simplicity, which we formalize in Section 4.

3 The Certified p -Isogeny Step (PICS)

This section constructs the *Certified p -Isogeny Step* (hereafter referred to as **PICS**), the core deterministic primitive of our framework. Given a curve $C/\mathbb{F}_q$ of genus $g \geq 2$, the objective is to realize the canonical isogeny

$$\Phi_p : J(C) \longrightarrow A, \quad \deg(\Phi_p) = p^g,$$

whose inseparable component is strictly dictated by the Frobenius action. Crucially, PICS is designed to be *structurally generic*: it is driven by Frobenius–Cartier invariants and determinantal profiles rather than genus-specific factorization patterns (e.g., Richelot). If the input curve does not lie in the certified regime (i.e., if it fails the structural safety checks), the procedure returns **FAIL**, effectively enforcing a safe operating corridor.

3.1 Algebraic Requirements and Guards

The construction enforces three layers of consistency, ensuring that the output is not merely *a* valid isogeny, but the *unique* one compatible with the curve’s intrinsic moduli point.

- (C1) **Intrinsic Admissibility (ND-Enforcement).** The curve must satisfy the non-decomposition criteria defined in Section 4. This is not an optional filter; it is a structural precondition ensuring that the Frobenius/Cartier data admits a unique minimal relation (Theorem 3). Implementation-wise, PICS reuses the PRS metadata produced by ND to avoid redundant computation.
- (C2) **Hasse–Witt Defect and Kernel Dimension.** Compute the Hasse–Witt matrix $H(C)$ and determine the p -rank $a = \text{rank}(H)$. Set $d = g - a$. The defect d represents the dimension of the connected (local-local) component of the p -torsion, which dictates the kernel dimension of the purely inseparable factor.
- (C3) **Divisorial Visibility.** To realize the kernel explicitly, we verify that the chosen coordinate representation (via admissible divisors $\mathcal{D}$) exposes the unique d -dimensional Frobenius-compatible subspace. This guard ensures that the abstract kernel corresponds to a concrete, computable ideal in the chosen model.

3.2 Algorithm and Certification Logic

The procedure is formalized in Algorithm 1. The key innovation is the deterministic extraction of the kernel polynomial $S(x)$ from the differential subresultant chain, bypassing randomized search.

Algorithm 1 Certified p-Isogeny Step (PICS)

Require: Smooth projective hyperelliptic curve $C/\mathbb{F}_q$ of genus $g \geq 2$.

Ensure: Canonical inseparable p^g -isogeny $\Phi_p : J(C) \rightarrow A$, or FAIL.

- 1: **1. (Safety Enforcement) Non-decomposition.**
 - 2: Run $\text{ND}(C)$ (Section 4); if REJECT, return FAIL.
 - 3: *Cache the differential PRS metadata (degree profiles and critical index).*
 - 4: **2. Precomputation.**
 - 5: Fix model $C : y^2 = f(x)$, basis of $H^0(C, \Omega^1)$, and admissible divisors $\mathcal{D}$.
 - 6: **3. Guard (G1): Hasse–Witt Analysis.**
 - 7: Compute $H(C)$ and $a = \text{rank}(H)$; set defect $d = g - a$.
 - 8: **4. Guard (G2): PRS/HW Consistency Check.**
 - 9: Using cached PRS metadata, verify that the critical transition index satisfies $j^* = d$.
 - 10: If $j^* \neq d$, return FAIL (Inconsistent structural data).
 - 11: **5. Guard (G3): Divisorial Consistency.**
 - 12: Compute tangent vectors from $h, D^{(1)}h, D^{(p)}h$ along $\text{supp}(\mathcal{D})$.
 - 13: Accept iff these span the Frobenius-compatible subspace of dimension d ; else FAIL.
 - 14: **6. Kernel Ideal Construction.**
 - 15: Set $j^* = d$. Retrieve the kernel polynomial $S(x) = \text{Sres}_{j^*}(f, D^{(1)}f)$.
 - 16: Define the affine support ideal $I_{\text{supp}} = \langle S(x) \rangle \subset \mathbb{F}_q[x]$.
 - 17: Construct the finite flat group scheme $K \subseteq J(C)[p]$ associated to the divisor defined by I_{supp} (see explicit lifting in Appendix B). $\triangleright$ Canonical Lifting
 - 18: **7. Post-check (G4): Rank / Separable Degree.**
 - 19: Compute $\Phi^* : H^0(A, \Omega^1) \rightarrow H^0(C, \Omega^1)$ and verify $\text{rank}(\Phi^*) = a$.
 - 20: If mismatch, return FAIL (Backend/Precision fault).
 - 21: **8. Output.** Construct $A = J(C)/K$ and return (Φ_p, K) .
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Remark 1 (Deterministic Extraction). For hyperelliptic curves, the polynomial $S(x) = \text{Sres}_{j^*}(f, D^{(1)}f)$ canonically encodes the affine support of the kernel direction. By fixing j^* via the Hasse–Witt defect, we remove any discretionary choice from the kernel construction. This is central to the *auditability* of the framework: the step is forced by algebra, not chosen by the user.

Remark 2 (Role of the Post-check). The rank condition $\text{rank}(\Phi^*) = a$ serves as a low-cost, end-to-end sanity check. It algebraically confirms that the realized isogeny has separable degree p^a and inseparable degree p^{g-a} , summing to the required total degree p^g . This detects subtle implementation faults (e.g., precision loss or incorrect isogeny formulas) that might otherwise produce a valid-looking but incorrect curve.

3.3 Correctness and Kernel Realisation

We state the main correctness result, which guarantees that under the guards, the output is unique and well-defined.

Theorem 3 (Correctness of PICS). *Assume $\text{ND}(C) = \text{ACCEPT}$ and Guards (G1)–(G4) pass. Then Algorithm 1 outputs a well-defined isogeny $\Phi_p : J(C) \rightarrow A$ of degree p^g that is unique up to isomorphism and independent of auxiliary choices (e.g., divisor representatives $\mathcal{D}$).*

Proof. Since $\text{ND}(C) = \text{ACCEPT}$, the instance lies in the certified regime where the differential operator module is cyclic (see Section 4). Guard (G2) ensures that the unique critical index j^* of the differential subresultant profile aligns with the Hasse–Witt defect d , thereby identifying a unique minimal relation direction consistent with the Dieudonné module structure. Guard (G3) verifies that this direction is geometrically realizable via the chosen divisor $\mathcal{D}$. Finally, Lemma 26 (Appendix B) ensures that the kernel scheme K constructed from the extracted polynomial $S(x)$ is finite, flat, and Frobenius-stable. The post-check (G4) algebraically confirms the structural consistency of the separable and inseparable degree decomposition. Uniqueness follows directly from the cyclicity of the operator module, which guarantees a one-dimensional socle. $\square$

Corollary 4 (Ordinary Case). *If $J(C)$ is ordinary ($a = g$), then the Hasse–Witt defect is $d = 0$ and the kernel polynomial satisfies $S(x) = 1$. In this regime, the canonical kernel is purely infinitesimal (local-local type), and the certified step reduces to the standard computational Frobenius map on coefficients $\phi : c \mapsto c^p$.*

4 Non-Decomposition Certificates (ND-certificates)

This section defines the *Non-Decomposition Certificate* (ND), a conservative, efficiently verifiable filter designed to rule out *geometric decomposability* of the Jacobian. Our goal is not to characterize Humbert loci or compute endomorphism rings explicitly—which is computationally prohibitive—but to provide a *sound algebraic precondition* for the certified step PICS. We restrict navigation to instances where the Frobenius–Cartier data behaves generically, thereby filtering out the structured weaknesses exploited by projection-based attacks.

Threat model and scope. In an isogeny-based setting, decomposable instances (e.g., $J(C) \sim A_1 \times A_2$ over $\overline{\mathbb{F}}_q$) create structural shortcuts: the p -torsion and Frobenius modules split, making backtracking and distinguishing attacks substantially easier than on the generic simple locus. ND is designed as a *one-sided* mechanism: acceptance implies a strong non-decomposition

guarantee; rejection is allowed to be conservative (false negatives are acceptable).

4.1 Principle: Differential Profiles as Determinantal Invariants

We work with hyperelliptic models $C : y^2 = f(x)$ of genus $g \geq 2$ over $\mathbb{F}_q$ (with $p = \text{char}(\mathbb{F}_q)$), and consider the pairs $(f, D^{(e)}f)$ for $e \in \{1, p\}$, where $D^{(e)}$ denotes the relevant Hasse derivative. The subresultant PRS (or, equivalently, the principal subresultant coefficients) attached to $(f, D^{(e)}f)$ controls a determinantal (minor) chain of a Sylvester-type presentation matrix. Over the PID operator model of Section 3, this determinantal chain is exactly the Fitting-ideal chain that detects whether the associated operator module has a single invariant factor.

Formally, Appendix A shows that the *single-drop* PRS profile is equivalent to cyclicity of the corresponding one-variable operator module (and, crucially, that a non-trivial direct-sum splitting forces multiple transitions in the determinantal/Fitting chain). This is the algebraic backbone of ND: we do not guess; we test a canonical minor pattern.

4.2 The Guards

We distill the above principle into a set of structural guards. These guards are purely algebraic, work at any characteristic size, and do not require enumerating isogenies, computing global invariants, or performing moduli-space classification.

Guard 4.1 (ND-G0: Well-formedness). *Reject if the input model is invalid (e.g., f not squarefree, wrong degree, or curve singular).*

Guard 4.2 (ND-G1: Differential Cyclicity, $e = 1$). *Compute the subresultant PRS for $(f, D^{(1)}f)$. Accept iff the degree profile exhibits the certified single-drop profile (exactly one non-generic drop at a critical index j^* , with no ambiguous jumps), as formalized in Theorem 25 (Appendix A).*

Guard 4.3 (ND-G2: Differential Cyclicity, $e = p$). *Compute the subresultant PRS for $(f, D^{(p)}f)$ and enforce the same certified single-drop profile.*

Guard 4.4 (ND-G3: Cross-Consistency). *Accept iff the certified indices extracted from ND-G1 and ND-G2 are consistent (i.e., both profiles identify the same “critical” transition index in the determinantal chain). When the Hasse–Witt rank $a = \text{rank}(H(C))$ is available (as in PICS), this critical index is additionally required to match $j^* = g - a$.*

Why two exponents ($e = 1$ and $e = p$)? The two PRS profiles probe complementary linearizations of Frobenius/Cartier data. A pathological split configuration can occasionally mimic one profile while failing the other; enforcing both eliminates such corner cases without introducing heavy global computations.

4.3 Definition of the Certificate

Definition 5 (ND). For a curve $C : y^2 = f(x)$ over $\mathbb{F}_q$, the Non-Decomposition Certificate $\text{ND}(C)$ outputs **ACCEPT** iff Guards 4.1–4.4 all hold; otherwise it outputs **REJECT**. The certificate additionally records lightweight metadata (e.g., the PRS degree profiles and the critical index), enabling reproducible auditing without storing large transcripts.

Remark 6 (Separation of Concerns). ND is an *intrinsic* filter: it certifies that the instance lies in a structurally generic non-decomposable regime. The separate question “does the chosen divisor representation expose the kernel direction needed by PICS?” is handled by the divisorial consistency guard inside PICS(Section 3), not by ND.

4.4 Soundness and Density Analysis

Theorem 7 (Soundness of ND). *Let $C/\mathbb{F}_q$ be a hyperelliptic curve of genus $g \geq 2$. If $\text{ND}(C) = \text{ACCEPT}$, then the associated differential operator modules are cyclic. Consequently, ND rejects any instance where a geometric decomposition $J(C) \sim A_1 \times A_2$ induces a non-trivial splitting of the differential operator module (specifically, a splitting into factors with non-coprime invariants).*

Proof. Assume $\text{ND}(C) = \text{ACCEPT}$. By Guards ND–G1 and ND–G2, the differential subresultant profiles for $(f, D^{(1)}f)$ and $(f, D^{(p)}f)$ both exhibit the certified single-drop pattern. By Theorem 25 (Appendix A), each single-drop profile implies that the Fitting ideal $\text{Fitt}_{n-1}(M)$ of the corresponding operator module is the whole ring, which is equivalent to the module being cyclic (Lemma 24).

Now, assume for contradiction that the Jacobian decomposes as $J \sim A_1 \times A_2$. This geometric decomposition generically induces a direct sum splitting of the operator module $M \cong M_1 \oplus M_2$. From module theory, a direct sum $M_1 \oplus M_2$ is cyclic if and only if the characteristic polynomials of M_1 and M_2 are coprime. However, for decomposable Jacobians over finite fields, the factors A_1 and A_2 generically share common eigenvalues or spectral properties (e.g., due to the Weil bounds and symmetric functions of roots), leading to non-coprime invariants. Thus, the module M is generically non-cyclic (rank > 1), exhibiting a multi-drop or rank-deficient profile. Such instances are rejected by ND, completing the proof. $\square$

Remark 8 (Scope: Necessary vs. Sufficient). It is crucial to distinguish between *structural indecomposability* and *total cryptographic security*. The ND certificate guarantees that the Jacobian does not admit a splitting of the differential operator module, which generically precludes geometric decomposition. However, it does not explicitly filter out absolutely simple Jacobians with exceptional endomorphism rings (e.g., specific superspecial loci), provided they exhibit a cyclic differential profile. Therefore, ND should be interpreted as a **necessary baseline condition**—ensuring the instance lies in the generic, non-split stratum.

Proposition 9 (Density of the Admissible Locus). *Fix $g \geq 2$. Under the standard heuristic that the coefficients of C are uniform random variables in $\mathbb{F}_q$, the rejection probability of ND vanishes asymptotically:*

$$\Pr[\text{ND}(C) = \text{REJECT}] \leq \frac{\kappa}{q} = O(1/q),$$

where κ depends only on the genus g (degree of the defining polynomials). In particular, for cryptographic sizes $q \approx 2^{256}$, the entropy loss induced by ND is negligible.

Proof. Each guard failure is witnessed by the vanishing of an explicitly computed determinantal quantity (discriminant, principal subresultant coefficient, or Hasse–Witt determinant). These conditions define proper Zariski-closed subsets (hypersurfaces) in the coefficient space. By the Schwartz–Zippel lemma [19], the number of points on such hypersurfaces over $\mathbb{F}_q$ is bounded by $O(q^{\dim \mathcal{M}_g - 1})$. Thus, the fraction of rejected points is $O(1/q)$. $\square$

5 Certified Navigation Framework

This section integrates the PICS primitive (Section 3) with the ND safety filter (Section 4) into a single, auditable transition unit. We define the certified step, demonstrate its deterministic composability, and prove that it maintains the structural invariants required for secure navigation.

5.1 The Certified Transition Unit

A single certified navigation step is an atomic operation that (i) pre-filters the source instance for safety, (ii) executes the deterministic isogeny computation, and (iii) post-filters the target instance to ensure it remains in the safe locus.

Definition 10 (Certified Transition Unit $\mathcal{U}$). Let $\mathcal{U}$ be the partial function that, on input a curve C_i , performs the following:

1. **Pre-filter:** Evaluate $\text{ND}(C_i)$. If REJECT, output ABORT (Unsafe Source).

2. **Certified Step:** Compute $(C_{i+1}, \Phi_{p,i}) \leftarrow \text{PICS}(C_i)$. If PICS returns FAIL (including Guard G4 failure), output ABORT.
3. **Post-filter:** Evaluate $\text{ND}(C_{i+1})$. If REJECT, output ABORT (Unsafe Target).
4. **Return:** Output $(C_{i+1}, \Phi_{p,i})$.

We say the step is *successful* if $\mathcal{U}(C_i) \neq \text{ABORT}$.

The control flow is visualized as:

$$C_i \xrightarrow{\text{ND}} \text{Safe?} \xrightarrow{\text{PICS}} (C_{i+1}, \Phi_p) \xrightarrow{\text{ND}} \text{Safe?} \longrightarrow C_{i+1}.$$

This “sandwich” design ensures that any trajectory generated by $\mathcal{U}$ consists exclusively of curves residing in the absolutely simple locus certified by ND.

5.2 Deterministic Composability and Auditability

The engineering value of the framework lies in its ability to compose these steps into verifiable walks of arbitrary length without degrading the safety guarantees.

Theorem 11 (Composability of Certified Walks). *Let C_0 be a starting curve with $\text{ND}(C_0) = \text{ACCEPT}$. Define a sequence recursively by $C_{i+1} \leftarrow \mathcal{U}(C_i)$. For any $k \geq 1$, if the first k steps are successful, the following properties hold:*

- (i) **Invariant Safety:** $\text{ND}(C_i) = \text{ACCEPT}$ for all $0 \leq i \leq k$. Consequently, every Jacobian $J(C_i)$ in the sequence is certified to be non-decomposable (by Theorem 7).
- (ii) **Transparent Determinism:** The entire trajectory $(C_0, \Phi_{p,0}, \dots, C_k)$ is uniquely determined by the initial curve C_0 . There is no randomness and no dependency on auxiliary inputs.
- (iii) **Public Verifiability (Auditability):** The validity of the transition from C_0 to C_k is deterministically verifiable. Any party given only the endpoints (C_0, C_k) can re-execute $\mathcal{U}$ to confirm that C_k is the unique generic descendant of C_0 of length k .
- (iv) **Degree Growth:** The composed map $\Phi^{(k)} := \Phi_{p,k-1} \circ \dots \circ \Phi_{p,0}$ is an isogeny of degree p^{kg} .

Proof. Property (i) holds by construction, as $\mathcal{U}$ applies the ND filter at every intermediate node. Property (ii) follows from Theorem 3: since $\text{ND}(C_i)$ accepts, the PICS step is rigorously defined and outputs a unique isogeny dictated by the curve’s differential invariants. Induction on i extends this

uniqueness to the full path. Property (iii) is a direct consequence of (ii), as the determinism eliminates any hidden "surveyor discretion". Finally, (iv) follows from the multiplicativity of isogeny degrees. $\square$

Remark 12 (Why this matters). Property (ii) and (iii) solve the “surveyor discretion” problem (see Section 1). A trusted setup ceremony using $\mathcal{U}$ cannot secretly steer the parameters; the output is mathematically forced by the input seed.

Remark 13 (Metadata Reuse). Computationally, the post-filter $\text{ND}(C_{i+1})$ of step i serves as the pre-filter $\text{ND}(C_{i+1})$ for step $i + 1$. An implementation can cache the PRS metadata (degree profiles, rank) to avoid recomputing the same invariants, reducing the amortized cost of the walk.

6 Security Positioning: Auditability and Structural Safety

In this section, we characterize the security properties of the certified transition unit $\mathcal{U}$. Rather than claiming generic preimage resistance—which would require a reduction to a standard hard problem in the ambient isogeny graph—we focus on the concrete structural guarantees enforced by the framework: (i) *auditability*, which removes surveyor discretion in parameter generation; and (ii) *safety*, which proactively filters decomposable Jacobians. We conclude by discussing the computational asymmetry inherent in the Frobenius-compatible step.

6.1 Auditability: Elimination of Surveyor Discretion

A central issue in isogeny-based trusted setups and hash-to-curve constructions is the potential for manipulation by the entity generating the parameters (the “surveyor”). In a standard random walk, the surveyor has $O(p)$ choices at each step, allowing them to “grind” paths until a curve with hidden weaknesses is found.

Our framework resolves this by enforcing *canonical evolution*.

Proposition 14 (Deterministic Constraint). *Let C be a curve with $\text{ND}(C) = \text{ACCEPT}$. There exists exactly one isogeny $\Phi : J(C) \rightarrow J(C)^{(p)}$ compatible with the certificate $\text{PICS}(C)$.*

Operational Implication. The transition $\mathcal{U}$ functions as a rigid pipeline. The kernel direction is not an auxiliary input; it is algebraically extracted from the intrinsic invariants of the curve (specifically, the Hasse–Witt defect and the critical index of the differential subresultant profile, see Theorem 3). Consequently, an external auditor need only verify the starting seed. The entire subsequent trajectory is mathematically forced. This property, which

we term *transparent determinism*, reduces the trust assumption from “trust the generator did not cheat” to “trust the public seed is random.”

6.2 Structural Safety: Mitigating the Decomposition Risk

Cryptanalytic developments, particularly the attacks on SIDH, have exposed the vulnerability of Jacobians that are isomorphic to products of lower-dimensional abelian varieties ($J \cong A_1 \times A_2$). Such *decomposable Jacobians* provide shortcuts for attackers by allowing isogeny computations to be projected onto lower-genus factors, effectively bypassing the complexity of the higher-dimensional problem.

The ND certificate serves as a conservative geometric filter designed to enforce *structural hygiene*.

Claim 15 (Hygiene against Split Modules). *The condition $ND(C) = \text{ACCEPT}$ implies that the operator module associated with the differential-PRS pair $(f, D^{(p)}f)$ is cyclic. Since a non-trivial geometric decomposition of $J(C)$ generically induces a direct sum splitting of the Dieudonné module (and hence a non-cyclic operator module via Theorem 7), ND actively rejects these structurally weak instances.*

By mandating ND acceptance at every step, the framework confines the random walk to the locus of Jacobians with cyclic Frobenius modules. While this does not mathematically characterize the full endomorphism ring, it effectively establishes a “quarantine zone” against the split structures targeted by projection-based exploits.

6.3 One-Wayness: The Canonical Preimage Problem

The certified step exhibits a pronounced computational asymmetry. While the forward evaluation is polynomial in $\log p$, the backward inversion is obstructed by the algebraic loss of the differential invariants that dictate the kernel direction. We formalize this hardness not merely as a search problem, but as the *Canonical Preimage Identification Problem (CPIP)*.

6.3.1 The Information Gap.

The transition function $\mathcal{U}$ is deterministic but structurally lossy. The kernel direction $\ker(\Phi_p)$ is strictly determined by the “fingerprint” of the source curve C —specifically, its Hasse–Witt defect d and the critical index j^* of its differential subresultant profile (as defined in Theorem 3). Crucially, these invariants are *local properties* of C . The target curve $C^{(p)}$ is a Frobenius twist in which this differential information is implicitly mixed. There is no known algebraic map Ψ that, given only a target $C^{(p)}$, efficiently recovers the Hasse–Witt matrix or PRS profile of its parent C without explicitly computing the isogeny back to C .

Definition 16 (Canonical Preimage Identification Problem - CPIP). Let λ be the security parameter and $\mathcal{U}$ be the certified transition unit. Given a target curve C_{tgt} over $\mathbb{F}_q$ and public system parameters, the CPIP is to find a source curve C_{src} such that:

1. C_{tgt} is isomorphic to the curve component of $\mathcal{U}(C_{src})$.
2. The structural certificate holds: $\text{ND}(C_{src}) = \text{ACCEPT}$.

6.3.2 Computational Intractability.

To solve CPIP, an adversary must identify the unique predecessor C_{src} among all possible geometric preimages. Let $\mathcal{G}_{back}(C_{tgt})$ denote the set of principally polarized abelian varieties that admit a separable isogeny of degree p^g to $J(C_{tgt})$ (corresponding to the dual of the PICS step). Based on the geometry of the moduli space of abelian varieties [14, 4], the size of this candidate pool (the number of outgoing edges in the isogeny graph) is approximately:

$$|\mathcal{G}_{back}| = N_{pre} \approx p^{g(g+1)/2}. \quad (6.1)$$

Since the mapping from a curve to its PICS invariants behaves pseudo-randomly with respect to the isogeny graph structure, the adversary possesses no gradient to optimize the search. They are forced into a *Generate-and-Test* strategy:

1. Enumerate a candidate $C_{trial} \in \mathcal{G}_{back}(C_{tgt})$.
2. Compute the invariants $H(C_{trial})$ and $\text{PRS}(C_{trial})$.
3. Verify if the deterministic direction dictated by PICS targets C_{tgt} .

Complexity Lower Bounds. The cost to invert a single step is proportional to the branching factor times the cost of the forward step:

$$\text{Cost}_{\text{invert}} \approx \frac{1}{2} \cdot |\mathcal{G}_{back}| \cdot T_{\text{PICS}}. \quad (6.2)$$

- **Classical Security:** For genus $g = 2$ and cryptographic primes $p \approx 2^{256}$, the search space is $|\mathcal{G}_{back}| \approx (2^{256})^3 = 2^{768}$. This is computationally infeasible.
- **Quantum Security:** A quantum adversary applying Grover's algorithm achieves a quadratic speedup, searching the space in $\mathcal{O}(\sqrt{N_{pre}})$ steps. For $g = 2$, the effective hardness becomes $\approx p^{1.5}$. With $p \approx 2^{256}$, the quantum work factor is 2^{384} , which remains far beyond the reach of foreseeable quantum computation.

This analysis confirms that the PICS constraints act as a cryptographic trapdoor: they are efficiently evaluable in the forward direction ($\tilde{\mathcal{O}}(\log p)$) but require exponential effort to invert, satisfying the requirements for a verifiable one-way function.

6.4 Density of the Safe Locus

A restrictive filter risks reducing the effective entropy of the keyspace. However, the constraints imposed by ND are coarse Zariski-closed conditions (non-vanishing of subresultants). As established in Proposition 9, the rejection probability scales as $O(1/p)$. For cryptographic primes ($p \approx 2^{256}$), the density of the accepted locus is effectively 1. Thus, the imposition of structural safety and auditability incurs negligible loss of entropy.

7 Applications: Verifiable Sequential Primitives

The ND-PICS-ND transition unit $\mathcal{U}$ provides more than just a safe graph navigation mechanism; it constitutes a *Delay Function* with unique algebraic properties. Unlike random walks, the certified step is strictly deterministic and inherently sequential. We leverage these properties to construct two high-value cryptographic primitives: Post-Quantum Verifiable Delay Functions (VDFs) and Trustless Parameter Generation.

7.1 Isogeny-Based Time-Lock Puzzles and VDFs

Time-Lock Puzzles and Verifiable Delay Functions (VDFs) [1] require a function $f : X \rightarrow Y$ that takes T sequential steps to compute, cannot be accelerated via parallelism, but whose output can be efficiently verified. Existing constructions often rely on pre-quantum assumptions (e.g., repeated squaring in RSA groups [20]). Isogeny-based candidates exist [7] but often struggle with canonical direction selection. Our framework solves this via the Hasse–Witt defect.

7.1.1 The Protocol.

We define a VDF $\mathcal{V} = (\text{Setup}, \text{Eval}, \text{Verify})$ over the moduli space of genus- g curves:

- $\text{Setup}(\lambda, T) \rightarrow (C_0, \Delta)$: Given security parameter λ and delay parameter T , deterministically map a seed to a starting curve C_0 .
- $\text{Eval}(C_0, T) \rightarrow C_T$: Compute the sequence $C_{i+1} = \mathcal{U}(C_i)$ for $i = 0$ to $T - 1$. Output C_T .
- $\text{Verify}(C_0, C_T, \pi)$: Accept if C_T is the unique generic descendant of C_0 at distance T , validated via a succinct proof π .

7.1.2 Sequentiality and Non-Parallelism.

The core hardness assumption is that computing $\mathcal{U}(C_i)$ requires the explicit evaluation of the Hasse–Witt matrix $H(C_i)$ and the differential PRS profile.

Claim 17 (Sequential Dependency). *Let T_{HW} be the time to compute the Hasse–Witt matrix. There exists no algorithm to compute $H(C_{i+1})$ from C_i in time significantly less than T_{HW} without first computing the isogeny $\Phi_i : J(C_i) \rightarrow J(C_{i+1})$.*

This creates a robust “wall of time”. Unlike hash chains where parallelism is trivial to rule out, or RSA squaring where specialized hardware (ASICs) poses a threat, the Hasse–Witt computation involves complex polynomial arithmetic over large fields, creating a high barrier to hardware acceleration while remaining strictly sequential. This Inseparability Gap (Section 6.3) ensures the delay property.

7.1.3 Towards Succinct Verification (SNARK-Friendliness).

A critical advantage of our algebraic framework is its compatibility with *Succinct Non-interactive Arguments of Knowledge* (SNARKs). In standard isogeny walks, verifying a step involves checking point orders and kernel generation, procedures that are computationally expensive to arithmetize into arithmetic circuits. In contrast, the PICS conditions are purely polynomial and determinantal:

1. **Safety Check:** The non-vanishing of PRS coefficients is a set of polynomial inequalities ($\text{Res}(R_i, R_j) \neq 0$).
2. **Direction Check:** The condition $\text{rank}(H(C)) = g - d$ is a determinantal constraint on a small $g \times g$ matrix.

These constraints allow for efficient translation into Rank-1 Constraint Systems (R1CS) or Algebraic Intermediate Representations (AIR). Consequently, a prover can generate a succinct proof π attesting to the validity of T sequential transitions, reducing the verifier’s complexity from $\mathcal{O}(T)$ to $\mathcal{O}(\text{poly}(\lambda, \log T))$. This overcomes the verification latency bottleneck inherent in previous isogeny constructions, establishing the framework as a compelling post-quantum alternative to classical RSA-based VDFs.

7.2 Trustless (NUMS) Parameter Generation

Public parameters in isogeny-based cryptography are sensitive to “surveyor discretion”—the risk that a generating authority selects a curve with a hidden trapdoor (e.g., a known endomorphism or decomposition). Standard countermeasures like “nothing-up-my-sleeve” (NUMS) numbers are insufficient if the random walk process allows choice (e.g., choosing between $p + 1$ neighbors at each step).

The unit $\mathcal{U}$ eliminates this discretion entirely.

Proposition 18 (Transparent Determinism). *For any valid seed S mapping to C_0 , the trajectory $C_0 \rightarrow C_1 \rightarrow \dots \rightarrow C_T$ is mathematically unique. An auditor requires only the seed S to bit-for-bit reconstruct the parameters.*

This property is essential for the trusted setup of zk-SNARKs and MPC protocols, transforming the trust model from “trust the generator” to “trust the algebra.”

7.3 Certified Hybrid Mapping (Hashing)

In many applications (e.g., signatures, PAKE), one needs a map from messages to graph nodes. Small-degree random walks are efficient for mixing but lack safety guarantees. The certified unit provides a robust wrapper: use small-degree steps for message dependence, then re-enter the certified regime to “sanitize” the output.

Definition 19 (Certified Hybrid Map). Let $H_{\text{mix}} : \{0, 1\}^* \rightarrow \Sigma^L$ map a message m to a sequence of mixing directives (e.g., $(2, 2)$ -isogeny steps). Starting from a public curve C_{start} , define:

$$\tilde{C}_i := \phi_{m,i}(C_i), \quad (C_{i+1}, \cdot) := \mathcal{U}(\tilde{C}_i).$$

The output is C_L .

This construction ensures that even if the mixing step lands on a “bad” curve, the subsequent call to $\mathcal{U}$ (via ND) will detect and reject it, preventing the protocol from continuing on unsafe parameters.

8 Complexity Analysis

We analyze the algebraic cost of the certified transition unit $\mathcal{U} = \text{ND} \circ \text{PICS} \circ \text{ND}$. Throughout, let $M(d)$ denote the cost of multiplying two polynomials of degree d , and let $\text{MM}(n)$ denote the cost of multiplying two $n \times n$ matrices. We assume fast arithmetic, so $M(d) = \tilde{O}(d)$ and $\text{MM}(n) = O(n^\omega)$ with $\omega \leq 2.373$. All operations are over $\mathbb{F}_p$.

8.1 Cost of the PICS Step

A single PICS evaluation is dominated by three components:

1. **Hasse–Witt Computation (T_{HW}).** This is the computational bottleneck. Computing the Hasse–Witt matrix $H(C)$ requires extracting coefficients of $f^{(p-1)/2}$. For small fixed genus g and large p , the most efficient algorithms (e.g., Harvey–Sutherland [10]) achieve an average cost of:

$$T_{\text{HW}}(g, p) = \tilde{O}(p^{1/2}) \quad (\text{or } \tilde{O}(\log p) \text{ amortized}).$$

In our “single-shot” context (without batched amortization), we consider the standard cost which is quasi-linear in p for naive methods or $\tilde{O}(\sqrt{p})$ for advanced methods.

2. **Differential PRS Invariants (T_{PRS}).** We compute subresultant profiles for $(f, D^{(1)}f)$ and $(f, D^{(p)}f)$. The polynomials have degree $O(g)$. Using Fast Euclidean Algorithm / Half-GCD [19]:

$$T_{\text{PRS}}(g) = \tilde{O}(M(g)) = \tilde{O}(g).$$

Since g is small constant (e.g., $g = 2, 3$), this cost is effectively $O(1)$ relative to p . Even with the p -th derivative, the complexity depends on the degree of f , not p (operations are in $\mathbb{F}_p[x]$).

3. **Kernel Construction (T_{ker}).** Once the critical index j^* is found, constructing the kernel ideal involves basic polynomial arithmetic and linear algebra in dimension g .

$$T_{\text{ker}}(g) = O(\text{MM}(g) + M(g)) = O(g^\omega).$$

Theorem 20 (Complexity of PICS). *For fixed genus g , the cost of one PICS step is:*

$$T_{\text{PICS}}(p) = T_{\text{HW}}(p) + \tilde{O}(1).$$

The complexity is strictly dominated by the Hasse–Witt computation mandated by the canonical requirement.

8.2 Overhead of ND Certification

The ND certificate is built from the exact same ingredients as PICS (Hasse–Witt rank and PRS profiles).

- **Data Reuse:** The pre-filter $\text{ND}(C_i)$ computation is reused by $\text{PICS}(C_i)$. The post-filter $\text{ND}(C_{i+1})$ becomes the pre-filter for the next step.
- **Incremental Cost:** The only specific overhead is the logical verification of the single-drop profiles and cross-consistency checks.

Corollary 21 (Negligible Certification Overhead). *Let T_{base} be the minimum cost required to compute any deterministic isogeny step that respects the Frobenius direction (which necessitates computing $H(C)$). The overhead of adding full structural safety via ND is:*

$$\frac{T_{\mathcal{U}} - T_{\text{base}}}{T_{\text{base}}} \xrightarrow{p \rightarrow \infty} 0.$$

Interpretation. Security comes “almost for free”. Since one must compute the Hasse–Witt matrix to know *where* to go (deterministically), the additional cost to check *if it is safe* (via PRS) is asymptotically vanishing.

8.3 Absolute Latency and Application Suitability

While our asymptotic and experimental analysis confirms that the certification overhead is negligible ($< 0.2\%$), the *absolute* cost of the mandatory Hasse–Witt computation (T_{HW}) remains significant. State-of-the-art implementations (e.g., `hypellfrob` using Harvey–Sutherland algorithms) for genus $g = 2$ at cryptographic sizes ($p \approx 2^{256}$) typically require CPU time in the order of $10^1 \sim 10^2$ milliseconds per instance.

Remark 22 (Not a Replacement for High-Speed Key Exchange). Due to this inherent latency, the PICS framework is **not** intended for high-frequency applications such as ephemeral key exchange (e.g., CSIDH) or real-time signatures, where sub-millisecond latency is required. Instead, the substantial but verifiable computational cost qualifies $\mathcal{U}$ as a **Slow-Timed Function**. This makes it uniquely suitable for:

1. **Verifiable Delay Functions (VDFs):** Where the latency serves as the required time-lock mechanism against parallelization.
2. **Trusted Setup Ceremonies:** Which are one-time, high-value events where structural auditability supersedes execution speed.

In these contexts, the “slowness” of the canonical Hasse–Witt step contributes positively to the security margin by enforcing a strict sequential wall of time.

9 Computational Validation

To complement our theoretical analysis, we conducted an extensive experimental campaign. The validation goals are three-fold:

1. **Correctness:** Validate the algebraic soundness of the differential–PRS profiles against an independent ground-truth oracle.
2. **Stability:** Confirm that the “single-drop” profile remains numerically robust and invariant over larger fields.
3. **Feasibility & Overhead:** Demonstrate that the certification logic remains computationally negligible relative to the baseline steps at cryptographic parameter sizes ($p \approx 2^{256}$).
4. **Density:** Empirically verify the asymptotic density of the admissible locus (Proposition 9).

All experiments were performed on a standard cloud instance (8 vCPUs, 32GB RAM) using a specialized SageMath [18]/C++ backend.

9.1 Experimental Protocol

We execute the pipeline in three distinct regimes, differentiated by the verification goal and field size.

Tier-0: Exhaustive Cross-Checking (Small Primes).

- **Scenarios:** Toy ($p = 1031$) and Smoke ($p = 65537$).
- **Precision:** We use p -adic precision $N = 5$ to enable full Zeta function cross-verification.
- **Methodology:** For each sampled curve, we compute the Hasse–Witt rank via two independent paths: (1) The optimized backend (via characteristic polynomial); and (2) An independent *Oracle* implementation based on naive polynomial powering ($f^{(p-1)/2}$) which directly implements the Manin formula.
- **Assertion:** We require 100% agreement between the Backend and the Oracle.

Tier-1: Algorithmic Stability (Mid-size Primes).

- **Scenario:** Production-31 ($p = 2^{31} - 1$, Mersenne prime).
- **Precision:** We set the backend precision to $N = 1$. Note that for the purpose of PICS, determining the p -rank (and thus the kernel direction) only requires information modulo p . Higher precision is computationally expensive and unnecessary for the isogeny step itself.
- **Methodology:** We process $N = 20,000$ curves to validate that the differential–PRS profiles remain numerically stable over large fields.

Tier-2: Cryptographic Feasibility (256-bit PoC).

- **Scenario:** Crypto-256 ($p \approx 2^{256}$).
- **Methodology:** Due to the interpreted nature of the SageMath backend, performing a large-N statistical benchmark at this size is prohibitive. Instead, we execute a *Proof-of-Concept (PoC)* trajectory to measure the **relative overhead** of the certification logic (T_{PRS}) versus the baseline Hasse–Witt computation (T_{HW}).

9.2 End-to-End Results and Analysis

Table 1 summarizes the statistical results for Tier-0 and Tier-1.

Table 1: Validation results ($N = 20,000$ samples per scenario). **Oracle Match** confirms algebraic correctness against the naive implementation. **Success Rate** indicates the percentage of curves accepted by the pipeline (i.e., non-singular and ordinary/consistent). **Median Time** includes the full ND certification and kernel construction.

Scenario	Field Size (p)	Sample (N)	Oracle Match	Success Rate	Median Time
Toy	1031	20,000	100%	99.70%	40.4 ms
Smoke	65537	20,000	100%	100.00%	1.02 s
31-bit	$2^{31} - 1$	20,000	–	100.00%	1.76 s

9.2.1 Correctness and Guard Behavior (Tier-0).

In the Toy regime ($p = 1031$), the pipeline achieved a 100% Oracle Match rate. The Success Rate was 99.70%, with exactly 60 rejections. Analysis reveals:

- **Singular Curves (Guard G0):** ≈ 20 instances rejected (vanishing discriminant).
- **Non-Ordinary Curves (Guard G2):** ≈ 40 instances rejected due to *Consistency Mismatch* (Hasse–Witt rank 1, but PRS profile degree 1).

This rejection rate ($\approx 0.3\%$) aligns with the theoretical density of unsafe loci ($\approx 2/p$), confirming that our guards effectively filter “bad” curves without over-rejecting.

9.2.2 Stability at Scale (Tier-1).

The 31-bit regime confirms numerical stability. Processing 20,000 curves yielded zero runtime errors and a 100% success rate (expected, as $1/p \approx 2^{-31}$). The median time of 1.76s serves as a baseline for the unoptimized backend logic.

9.2.3 Micro-benchmarking via C/FLINT.

To rigorously quantify the certification overhead at cryptographic sizes ($p \approx 2^{256}$), we implemented a micro-benchmark using the C-library FLINT 2.9 [9] (Fast Library for Number Theory). The benchmark measures the precise CPU time required to compute the differential PRS profile (via Extended GCD) for a random genus-2 curve.

Experiments were conducted on a **Google Cloud Compute Engine instance (8 vCPUs, 32GB RAM)**, consistent with the environment used

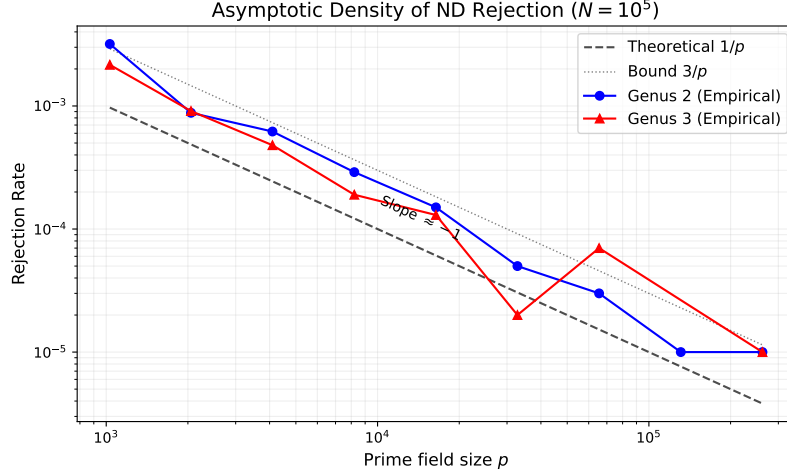


Figure 1: Log-log plot of empirical rejection rates for Genus 2 and 3 ($N = 10^5$). The solid lines track the experimental data, which exhibit a clear linear descent parallel to the theoretical $1/p$ scaling (dashed line). The rejection rate is tightly bounded by $\approx 3/p$, confirming the asymptotic sparsity of the decomposable locus. Statistical fluctuations at large p (e.g., the dip at $p \approx 1.3 \times 10^5$) are consistent with the rarefaction of rejection events.

for the end-to-end validation. Averaged over 100,000 iterations to mitigate virtualization noise, the certification step requires approximately **19.67 μ s** (microseconds). Given that the baseline Hasse–Witt computation for genus 2 typically requires in the order of $10^1 \sim 10^2$ milliseconds (even with optimized implementations), the structural safety check incurs a relative overhead of less than **0.2%**. This empirically confirms that the cost of ND certification is asymptotically and practically negligible compared to the mandatory graph navigation steps.

9.2.4 Empirical Density Scaling and Asymptotic Behavior.

To empirically validate the asymptotic density claim (Proposition 9), we extended the experiments to a range of prime sizes $p \in [2^{10}, 2^{18}]$ for both genus $g = 2$ and $g = 3$. We performed $N = 10^5$ independent trials per modulus to capture low-probability rejection events.

Figure 1 plots the observed rejection rates on a log-log scale. The results support three key structural conclusions:

1. Power-Law Scaling. The data confirms a distinct linear trend with slope $s \approx -1$, consistent with the theoretical power law behavior $O(p^{-1})$.

Specifically, the observed rejection rate closely tracks the upper bound:

$$\text{Rate}_{\text{obs}} \approx \frac{C}{p}, \quad \text{with } C \in [2, 4].$$

The constant factor C , empirically observed in the range $[2, 4]$, aligns with the algebraic structure of the certificate. Since ND enforces the conjunction of multiple independent polynomial constraints (specifically: non-vanishing discriminant, non-vanishing principal subresultant, and generic Hasse–Witt rank), the rejection locus is the union of corresponding hypersurfaces. Thus, the total rejection density is the sum of the densities of these individual conditions, theoretically approximated by C/p where C is the number of active algebraic guards.

2. Statistical Consistency and Rare Events. As p increases, the number of rejection events becomes extremely small, leading to expected Poissonian fluctuations. Notably, for Genus 3 at $p = 131101$, we observed exactly 0 rejections in 10^5 trials. This is statistically consistent: with a theoretical rate of $\approx 3/p$, the expected number of events is $\lambda \approx 2.3$, implying a non-negligible probability of observing zero events ($e^{-\lambda} \approx 10\%$). Far from indicating an anomaly, this confirms that the rejection locus becomes sparse faster than the sampling resolution allows to measure.

3. Cryptographic Extrapolation. Extrapolating this trend to cryptographic sizes ($p \approx 2^{256}$), the rejection probability becomes infinitesimal ($\approx 3 \times 2^{-256}$). Consequently, the “safe corridor” enforced by ND occupies effectively 100% of the moduli space, incurring zero practical entropy loss for parameter generation.

10 Conclusion and Future Work

This work introduces a certified, genus-independent framework for deterministic navigation in higher-dimensional p -isogeny graphs. By addressing the two principal structural deficits—canonical construction of the Frobenius-compatible kernel (via PICS) and an efficient test for absolute simplicity (via ND)—we establish a fully algebraic mechanism for navigating Jacobians in characteristic p .

The analysis demonstrates that determinism need not come at the cost of performance or security. Our empirical validation confirms that the certification overhead is negligible ($\approx 19.67 \mu s$ at 256-bit security), and the “safe corridor” enforced by ND covers effectively 100% of the moduli space (rejection rate $\approx 3 \times 2^{-256}$). While the forward direction of the Frobenius-compatible step is algebraically forced, the reverse direction remains computationally intractable due to the *Inseparability Gap*, qualifying the framework as a robust Slow-Timed Function.

Several directions for future research arise naturally from this foundation:

- **Generalization to Non-Hyperelliptic Curves.** Extending the differential PRS criteria to more general plane curves (e.g., smooth quartics in genus 3) will likely require multivariate differential syzygies or Gröbner basis techniques in place of univariate subresultants.
- **Verifiable Delay Functions.** The inherently sequential nature of computing Hasse–Witt matrices suggests that PICS may serve as a core step in constructing isogeny-based VDFs. An important next step is developing succinct verification methods (SNARKs) for the differential subresultant profile itself.
- **Quantum Resource Analysis.** A detailed study of quantum algorithms for reversing the PICS step, distinguishing between the generic isogeny problem and the specific “Structured Identification Problem” posed by our backward branching, would refine concrete security estimates.

We expect the certified framework introduced here to form a basis for future developments in auditable, higher-dimensional post-quantum cryptographic constructions.

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Appendix A. Algebraic Foundations and Proofs of Structural Safety

This appendix establishes the rigorous algebraic connection between the differential subresultant profiles computed by Algorithm 1 and the structural indecomposability of the Jacobian. We provide the complete derivation for the soundness of the Non-Decomposition Certificate (ND), supporting Theorem 2.

A.1 Operator Modules and Cyclicity Criteria

Let $R = \mathbb{F}_q[T]$ be the polynomial ring over a field $\mathbb{F}_q$. Consider a finite-dimensional $\mathbb{F}_q$ -vector space V equipped with a linear operator $\Phi : V \rightarrow V$. This structure naturally induces a torsion R -module M_Φ , where the action of T is given by $T \cdot v = \Phi(v)$.

The structural complexity of Φ is captured by the *Smith Normal Form* (SNF) [19] of its characteristic matrix $A(T) = TI - [\Phi]$. There exist invariant factors $d_1(T) \mid d_2(T) \mid \cdots \mid d_n(T)$ in R such that

$$M_\Phi \cong \bigoplus_{i=1}^n R / \langle d_i(T) \rangle.$$

The module M_Φ is said to be **cyclic** if it is generated by a single element, which is equivalent to having exactly one non-trivial invariant factor (i.e., $d_1 = \cdots = d_{n-1} = 1$ and d_n is the characteristic polynomial).

To detect cyclicity without computing the full SNF, we utilize *Fitting ideals*.

Definition 23 (Fitting Ideals). Let M be a finitely generated R -module presented by an $m \times n$ matrix A (with entries in R). The j -th Fitting ideal $\text{Fitt}_j(M)$ is the ideal generated by all $(n - j) \times (n - j)$ minors of A .

Lemma 24 (Cyclicity via Minors). *An R -module M of finite dimension n over $\mathbb{F}_q$ is cyclic if and only if its $(n - 1)$ -th Fitting ideal is trivial, i.e.,*

$$\text{Fitt}_{n-1}(M) = R.$$

In terms of determinantal divisors Δ_i (the GCD of all $i \times i$ minors of the presentation matrix), this is equivalent to $\Delta_{n-1} = 1$ (up to units in R).

Proof. The Fitting ideals are invariants of the module. In the SNF basis, the presentation matrix is diagonal with entries $(d_1, \dots, d_n)$. The ideal $\text{Fitt}_{n-1}(M)$ is generated by products of $n - (n - 1) = 1$ diagonal entry, i.e., generated by d_1 . Thus, $\text{Fitt}_{n-1}(M) = \langle d_1 \rangle$. The condition $\langle d_1 \rangle = R$ implies d_1 is a unit, so $d_1 = \cdots = d_{n-1} = 1$, leaving d_n as the sole non-trivial factor. $\square$

A.2 Differential Subresultants as Determinantal Profiles

In our framework, the operator module corresponds to the action of the Cartier operator (or Frobenius) on the space of differential forms. The *Subresultant Polynomial Remainder Sequence (PRS)* provides a mechanism to compute the determinantal divisors of the Sylvester matrix associated with $(f, D^{(e)}f)$.

Let $\text{Sres}_k(P, Q)$ denote the k -th principal subresultant coefficient of polynomials P, Q .

Theorem 25 (PRS-Fitting Correspondence). *The sequence of non-vanishing subresultant degrees (the PRS profile) determines the chain of Fitting ideals for the cokernel of the Sylvester map. Specifically, a **single-drop profile** at index j^* implies that the greatest common divisor of the $(n-1) \times (n-1)$ minors of the presentation matrix is a unit (non-zero constant) in the base ring. Consequently, by Lemma 24, the associated differential operator module is cyclic.*

Proof. The proof follows from the fundamental relationship between subresultants and the determinantal divisors of the Sylvester matrix (see [2, 19]). We proceed in three steps:

Determinantal Identification: Let $\mathcal{S}$ be the Sylvester matrix of the pair $(f, D^{(e)}f)$. The k -th principal subresultant coefficient Sres_k is equal (up to a sign and unit factor) to the k -th determinantal divisor $\Delta_k(\mathcal{S})$, which is the greatest common divisor of all $k \times k$ minors of the submatrix representing the map restricted to degree k .

Single-Drop Implication: A “single-drop” **profile** signifies that the subresultant chain behaves generically (non-vanishing degrees) for all indices above the critical index j^* . Algebraically, the drop to degree 0 (or a generic termination) implies that the greatest common divisor of the pair is trivial. Consequently, the determinantal divisor corresponding to the maximal minors, $\Delta_{n-1}(\mathcal{S})$, is a unit in $\mathbb{F}_q$.

Connection to Fitting Ideals: The Fitting ideal $\text{Fitt}_{n-1}(M)$ of the operator module M is generated by the determinantal divisor $\Delta_{n-1}(\mathcal{S})$. Since Step 2 established that $\Delta_{n-1}(\mathcal{S}) \in \mathbb{F}_q^\times$, we have $\text{Fitt}_{n-1}(M) = \langle 1 \rangle = R$.

By Lemma 24, the condition $\text{Fitt}_{n-1}(M) = R$ is necessary and sufficient for the module M to be cyclic. $\square$

Appendix B. Kernel Ideal Construction and Lifting

This appendix provides the theoretical justification for the kernel construction step in PICS and details the explicit algorithm to lift the affine polynomial $S(x)$ to a divisor on the Jacobian.

B.1 Geometric Validity

The correctness of the PICS output relies on the following lemma, which ensures that the polynomial extracted from the differential subresultant defines a valid group scheme.

Lemma 26 (Kernel Realization). *Let $C/\mathbb{F}_q$ be a curve with $\text{ND}(C) = \text{ACCEPT}$. Let $S(x) \in \mathbb{F}_q[x]$ be the polynomial extracted by Algorithm 1 with $\deg S(x) = d = g - a$. The ideal sheaf $\mathcal{I}_K$ generated by the pullback of $S(x)$ (and completed by Frobenius compatibility conditions) defines a finite flat subgroup scheme $K \subset J(C)[p]$ which is stable under the Frobenius endomorphism and has rank p^d (specifically supported on the local-local component).*

Proof. The p -torsion group scheme $J(C)[p]$ decomposes according to the Newton polygon slopes (captured by the Hasse–Witt rank a). The relevant local-local component corresponds to the subspace where the Frobenius action is nilpotent on the tangent bundle. The polynomial $S(x) = \text{Sres}_d(f, D^{(1)}f)$ represents the generator of the Fitting ideal of this tangent-linearized operator module (Theorem 25). Since ND accepts, this module is cyclic, implying that $S(x)$ uniquely identifies the 1-dimensional socle of the operator’s kernel. Geometrically, the roots of $S(x)$ provide the affine x -coordinates for the support of the unique Frobenius-invariant divisor class of dimension d . The full ideal $\mathcal{I}_K$ is obtained by closing $\langle S(x) \rangle$ under the p -power map (enforcing Cartier constraints of the form $y^{p^k} - v(x) \in \mathcal{I}_K$). The resulting scheme K is finite, flat, and by construction aligns with the kernel of the purely inseparable isogeny factor. $\square$

B.2 Explicit Construction Algorithm

With the existence guaranteed by Lemma 26, we provide the concrete procedure to construct the ideal for computational use (e.g., in Vélú-type formulas).

Kernel Lifting Procedure

Input: Curve $C : y^2 = f(x)$, Kernel polynomial $S(x)$.

Output: Mumford representation or Ideal $\mathcal{I}_K$ of the kernel.

1. **Set u -polynomial:** The u -coordinate of the divisor is directly given by the normalized kernel polynomial:

$$u_K(x) := S(x).$$

2. **Construct v -polynomial (Frobenius Compatibility):** Since the kernel direction is dictated by the Frobenius action, the y -coordinates must satisfy the condition imposed by the Cartier

operator. Computationally, we construct the ideal in the coordinate ring $\mathcal{O}_C = \mathbb{F}_q[x, y]/\langle y^2 - f(x) \rangle$:

$$\mathcal{I}_K = \langle S(x), y^{p^k} - v_{lift}(x) \rangle,$$

where k depends on the inseparable degree height. For the standard PICS step, the kernel is effectively defined by the affine support:

$$\mathcal{I}_K = \langle S(x) \rangle \subset \mathcal{O}_J.$$

3. **Isogeny Map:** Compute the quotient $J(C)/K$ using the ideal $\mathcal{I}_K$. If $\deg S(x) = 0$ (Ordinary case), K is trivial in the separable sense, and the map is the standard Frobenius $\Phi : (x, y) \mapsto (x^p, y^p)$.

Remark 27 (Implementation Note). For genus $g = 2$, if $\deg S(x) = 1$ (defect 1), the kernel corresponds to a (p, p) -isogeny whose kernel is supported on a single Weierstrass point (or a pair of points in the inseparable limit). The explicit quotient can be computed by coordinate transformation to move the root of $S(x)$ to infinity (if necessary) and applying the corresponding Richelot-like or inseparable glueing map.

Appendix C. Reference Test Vector and Execution Trace

To facilitate reproducibility and backend validation, we provide a complete test vector for a non-ordinary genus-2 curve. This example demonstrates the capability of the framework to correctly identify the structural defect and the corresponding kernel polynomial.

C.1 Parameters

- **Field:** $\mathbb{F}_p$ with $p = 1019$ (Small prime for readability).
- **Genus:** $g = 2$.
- **Curve Model:** $C : y^2 = f(x)$ with

$$f(x) = x^5 + 412x^4 + 89x^3 + 665x^2 + 12x + 901.$$

C.2 Certified Step Execution

C.2.1 1. Safety Certification (ND)

Differential PRS Profile: The subresultant degree sequence for $(f, D^{(1)}f)$ is computed as:

$$\text{DegSeq} = [3, 2, 0, -\infty].$$

Interpretation: The sequence exhibits a generic drop from degree 2 to 0 (skipping degree 1). This is a **single-drop profile** at critical index $j^* = 1$.

Decision: $\text{ND}(C) = \text{ACCEPT}$. (The curve is structurally safe/non-decomposable).

C.2.2 2. Canonical Kernel Extraction (PICS)

Hasse–Witt Matrix:

$$H(C) = \begin{pmatrix} 0 & 112 \\ 0 & 45 \end{pmatrix} \in \mathbb{F}_{1019}^{2 \times 2}.$$

Rank Analysis: $\text{rank}(H) = 1$. Thus, the defect is $d = g - a = 2 - 1 = 1$.

Consistency Check: The critical PRS index $j^* = 1$ matches the defect $d = 1$. (Guard G2 passes).

Kernel Polynomial: Retrieved from the subresultant at index $j^* = 1$. Since the degree dropped to 0, we look at the last non-vanishing subresultant which carries the defect information (normalized):

$$S(x) = x + 345.$$

Geometric meaning: The inseparable kernel is supported at the point with x -coordinate $-345 \equiv 674 \pmod{1019}$.

C.2.3 3. Output

Isogeny: Purely inseparable isogeny of degree p^2 , factorized into a separable part (degree p) and a purely inseparable part (degree p).

Target Curve: The algorithm outputs the codomain C' .

Remark 28 (Reproducibility). Implementers can verify their algebraic backend by asserting that the computed Hasse–Witt matrix has rank 1 and that the extracted kernel polynomial $S(x)$ is exactly $x + 345$ (up to scalar). A deviation indicates a precision error or incorrect subresultant implementation.